Technology Stack

Date	10 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

The **Voice-Based Concept Understanding Analyser (VBCUA)** is developed using a modern technology stack that supports Artificial Intelligence, speech processing, semantic analysis, data visualization, report generation, and database management. The selected technologies ensure high performance, maintainability, scalability, and an interactive user experience.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*Streamlit, HTML, CSS*	*Provides an interactive and responsive web interface for uploading audio, displaying analysis results, dashboards, and reports with minimal development effort.*

2	**Backend / Server-Side**	*Python 3.14*	*Handles application logic, AI processing, speech transcription, semantic analysis, scoring, feedback generation, and report creation efficiently.*
3	**Database / Data Storage**	*SQLite*	*Lightweight relational database used to store transcripts, similarity scores, reports, analysis history, and user results without requiring a dedicated database server.*
4	**Cloud / Hosting / Deployment**	*Streamlit Community Cloud (Planned)*	*Enables easy deployment of the application for online access, demonstration, and future scalability.*
5	**Version Control & CI/CD**	*Git & GitHub*	*Used for source code management, version control, collaboration, backup, and maintaining project history.*
6	**Third-Party APIs / Other Tools**	*Faster Whisper, Sentence-BERT, Librosa, ReportLab, Plotly, Matplotlib*	*Faster Whisper converts speech to text, Sentence-BERT performs semantic similarity analysis, Librosa extracts audio features, ReportLab generates PDF reports, while Plotly and Matplotlib provide data visualization and analytics.*